

Day 1 :- Introduction to Networking Hardware Theory : Overview of modems, routers, switches, access points, and network cables.

Tasks 1 :- Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Ans. :-



1. Internet :-

- The Internet connection comes from the Internet Service Provider (ISP).

2. Modem :-

- The modem receives the Internet signal from the ISP.
- It converts the signal into a form that the home network can use.

3. Router :-

- The router connects to the modem.
- It creates the home network.
- It assigns IP addresses to devices using DHCP.

- It provides security through a firewall and allows multiple devices to share the Internet connection.

4. Switch :-

- A switch increases the number of wired Ethernet ports.
- It connects wired devices such as desktop computers, printers, TVs, or access points.

5. Access Point (AP) :-

- The access point provides or extends Wi-Fi coverage.
- It connects to the switch (or sometimes directly to the router) using an Ethernet cable.
- Devices connect wirelessly to the access point.

6. Laptop :-

- The laptop connects to the network through Wi-Fi.
- It can access the Internet and communicate with other devices on the network.

7. Smartphone :-

- The smartphone connects wirelessly to the Wi-Fi network.
- It can browse the Internet and use network services.

Tasks 2 :-List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Ans. :-

Networking Hardware	Main Function	Real-Life Example
Modem	Connects your home or office network to the Internet through the Internet Service Provider (ISP).	At home: A JioFiber or Airtel modem connects your house to the Internet.
Router	Shares the Internet connection with multiple devices and manages network traffic.	At home: A Wi-Fi router lets your laptop, smartphone, and smart TV use the same Internet connection.
Switch	Connects multiple wired devices within the same local network (LAN).	In an office or computer lab: Connects several desktop computers and printers using Ethernet cables.
Access Point (AP)	Provides or extends Wi-Fi coverage so wireless devices can connect to the network.	In a school, college, hotel, or large office: An access point extends Wi-Fi to areas where the router's signal is weak.
Network Cable (Ethernet Cable)	Transfers data between networking devices using a wired connection.	At home or in an office: Connects the router to a computer, switch, or access point for a fast and stable connection.

Tasks 3 :- Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Ans. :-

Ethernet Cable (Cat5e/Cat6) :-



Ethernet Cable :-An Ethernet cable is the most common network cable used in homes, schools, and offices.

It has RJ45 connectors at both ends.

It connects devices directly to a network for a fast and stable Internet connection.

Used :- Connecting a computer to a router, router to a switch , switch to an access point and Connecting printers, gaming consoles, and smart TVs.

Coaxial Cable :-



A coaxial cable has a single copper wire in the center surrounded by insulation and a protective metal shield.

It is designed to carry high-frequency electrical signals with minimal interference.

Used :- Cable TV connections , Internet connections, Connecting a cable modem to the ISP's network, CCTV camera systems ect.

Fiber Optic Cable :-



A fiber optic cable contains very thin strands of glass or plastic. It transmits data as light signals instead of electrical signals.

Used :- High-speed fiber Internet (FTTH) , Data centers,

Telecommunications networks, Undersea Internet cables connecting countries ect.

Tasks 4 :- Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Ans. :-

Comparison Table: Router vs. Switch :-

Feature	Router	Switch
Main Use	Connects different networks (such as a home network to the Internet) and shares the Internet among multiple devices.	Connects multiple devices within the same Local Area Network (LAN).
Typical Location	Found in homes, offices, schools, and businesses near the Internet connection (modem).	Found in offices, computer labs, schools, and data centers where many wired devices need to be connected.
How It Handles Data	Uses IP addresses to determine the best path and forward data between different networks.	Uses MAC addresses to send data directly to the correct device within the same network.
Network Connection	Connects one network to another (LAN to WAN/Internet).	Connects devices within the same network (LAN).
Internet Sharing	Yes, it shares one Internet connection with many devices.	No, it only connects devices on the local network.
Wireless Support	Most modern routers include built-in Wi-Fi for wireless devices.	A standard switch does not provide Wi-Fi; it only supports wired Ethernet connections.
Number of Ports	Usually has 4–8 LAN ports and 1 WAN port.	Available with many ports such as 8, 16, 24, or 48 Ethernet ports.
Example	A Wi-Fi router at home connecting laptops, smartphones, and smart TVs to the Internet.	A switch in a school computer lab connecting many desktop computers using Ethernet cables.